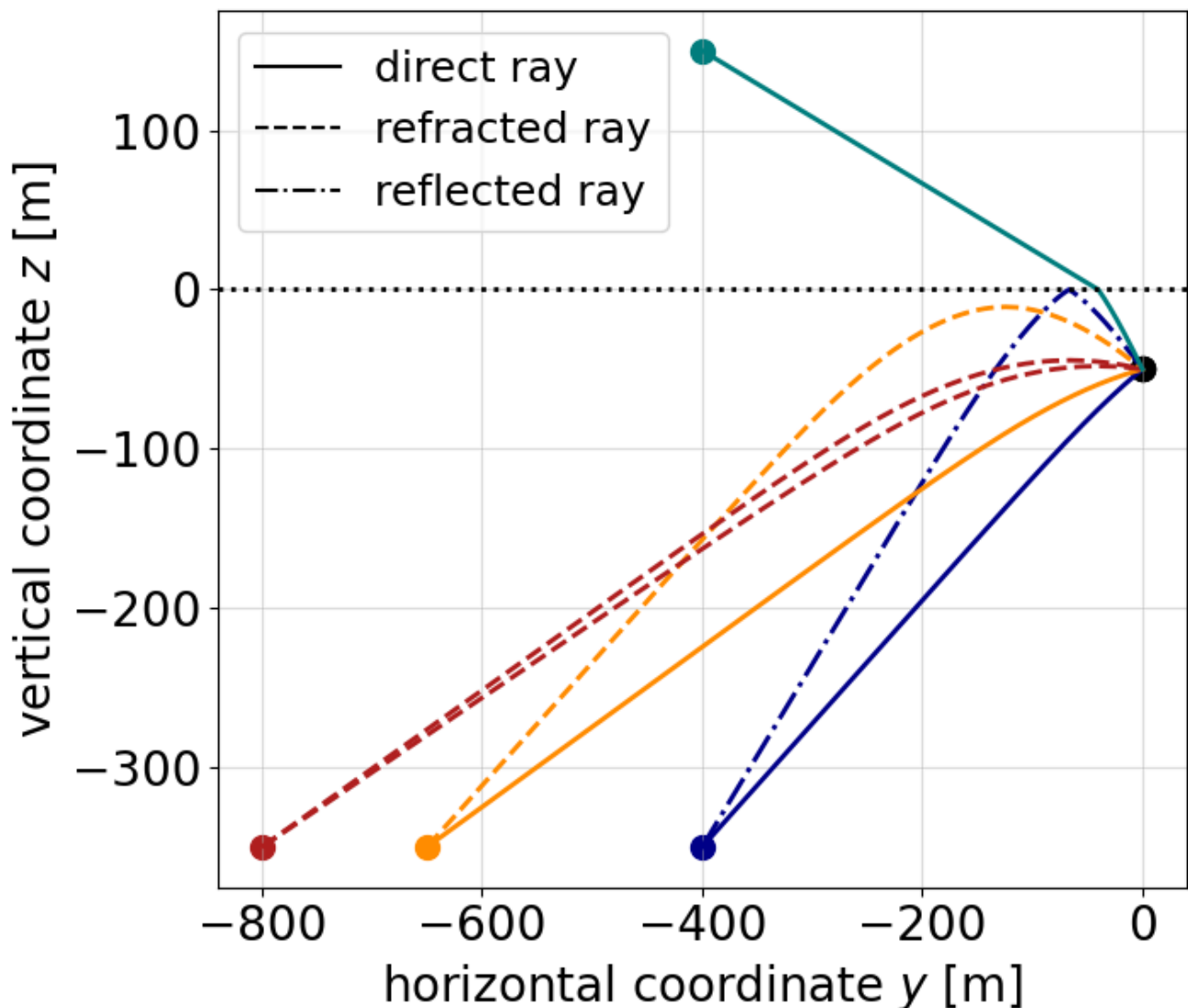


Ray-Tracing

This note is meant to collect useful information about analytic ray tracing in NuRadioMC. It was produced for private usage and is just provided here to maybe help in case someone wants to understand the fundamental principles of analytic ray tracing. It does not claim to be complete or entirely accurate. See the literature list at the end for citable sources. For questions feel free to contact hannes.warnhofer@desy.de

In ice the refractive index is not constant but depends on the depth, causing the signal paths do not follow a straight line but to get bent in shallower regions, where the refractive index changes the strongest. Besides being reflected at the surface, upwards going rays can have a turning point before reaching the surface and then turn downgoing again, arriving at the second point coming from above. In this case (turning point before reaching the receiver point) we call those rays refracted. If no turning point is present we call them direct and if they get reflected at the ice surface, well... The following plot shows an overview about the expected types of ray paths.



Deriving analytical solution from Fermat's principle

- Fermat's principle: Optical path between two points is the one where the travel time is minimal. Since we can express the propagation speed in a dielectric medium by $v = c \cdot n$, where n is the refractive index, this principle can be written mathematically as follows:

$$S = \int_A^B n \, ds \quad \text{and} \quad \delta S = 0$$

- If we assume a horizontally uniform refractive index which only depends on the depth coordinate, meaning $n = n(z)$, we can simplify the 3D problem to a 2D vertical plane that our signal propagation is taking place in. Based on this symmetry of $n(z)$ we can rotate the coordinate system so that $\dot{x} = dx/dz = 0$ and rewrite δS :

$$\begin{aligned} ds &= \sqrt{dx^2 + dy^2 + dz^2} = dz \sqrt{0 + (dy/dz)^2 + 1} = dz \sqrt{\dot{y}^2 + 1} \\ \Rightarrow \delta \int_A^B n(z) \sqrt{1 + \dot{y}^2} \, dz &= 0 \end{aligned}$$

- With a variable change we can rearrange and integrate this to the following:

$$\ln \dot{y} - \frac{1}{2} \ln (\dot{y}^2 + 1) = -\ln n + \text{constant}$$

which can also be written as:

$$\frac{\partial y / \partial z}{\sqrt{(\partial y / \partial z)^2 + 1}} n(z) = \frac{1}{C_0} = \text{constant}$$

- Notice how this shows already that there is a physical invariant in this problem, which we find as our integration constant C_0
- If we think about what the change $\partial y / \partial z$ means geometrically, we see that it describes at which angle (with respect to the vertical or horizontal) the signal propagates at a given point:

$$\tan (\partial y / \partial z)(z) = \theta(z)$$

- With this and some trigonometry relations we can rewrite our differential equation to a local snell's law:

$$\sin (\theta(z)) \cdot n(z) = \frac{1}{C_0} = \text{constant}$$

- We can define an approximate model for the refractive index that allows us to solve this equation analytically by exploiting the analytical properties of the exponential function:

$$n(z) = n_{\text{ice}} - \Delta_n \exp(z/z_0)$$

- For such an exponential refractive index model, the solution for the horizontal displacement y as a function of depth z can be found. With $\gamma = \Delta_n \exp(z/z_0)$:

$$y(z) = \pm C_0^{-1} c^{-1/2} z_0 \ln(F(\gamma)) \mp z_0 C_1'$$

where

$$F(\gamma) = \frac{\gamma}{2\sqrt{c(\gamma^2 - 2\gamma n_{\text{ice}} + c)} - 2\gamma n_{\text{ice}} + 2c}$$

- For our task it can help to think of this solution in the way that the displacement can be found from

$$y(z) = Y(z, C_0) + C_1$$

We have now found a mathematical solution that satisfies Fermat's principle and therefore describes the path that takes the least time for a signal propagating from a given point into a given direction for the a refractive index profile that follows the exponential model from above. However, as you will see in the plots below, this function is only physically valid up to the turning point, where the light propagation would become vertical.

The upper branches to this solution are not actual ray paths but a consequence of the depth dependent refractive index profile and our assumptions of rotational freedom, which allowed us to derive the depth dependent function $y(z)$.

To describe how an upwards going light signal propagates and possibly turns downwards at some point we can simply assume that after the turning point it follows the symmetric path it took on the way up. This means that the $y(z)$ solutions nevertheless gives us the information for the overall ray path by exploiting the symmetry, however one has to be careful to not run into artifacts based on the solution branches above the turning point when trying to find a ray tracing solution between two points by comparing the found $y(z_{\text{test}}, C_0)$ to y_{test} .

From $y(z)$ to finding ray paths

With this solution, we can find the path that a light ray takes under given initial conditions that are encoded in the two integration constants C_0 and C_1 , which simply acts as an offset parameter that shifts the whole solution horizontally. The parameter C_0 defines the shape or direction of the ray.

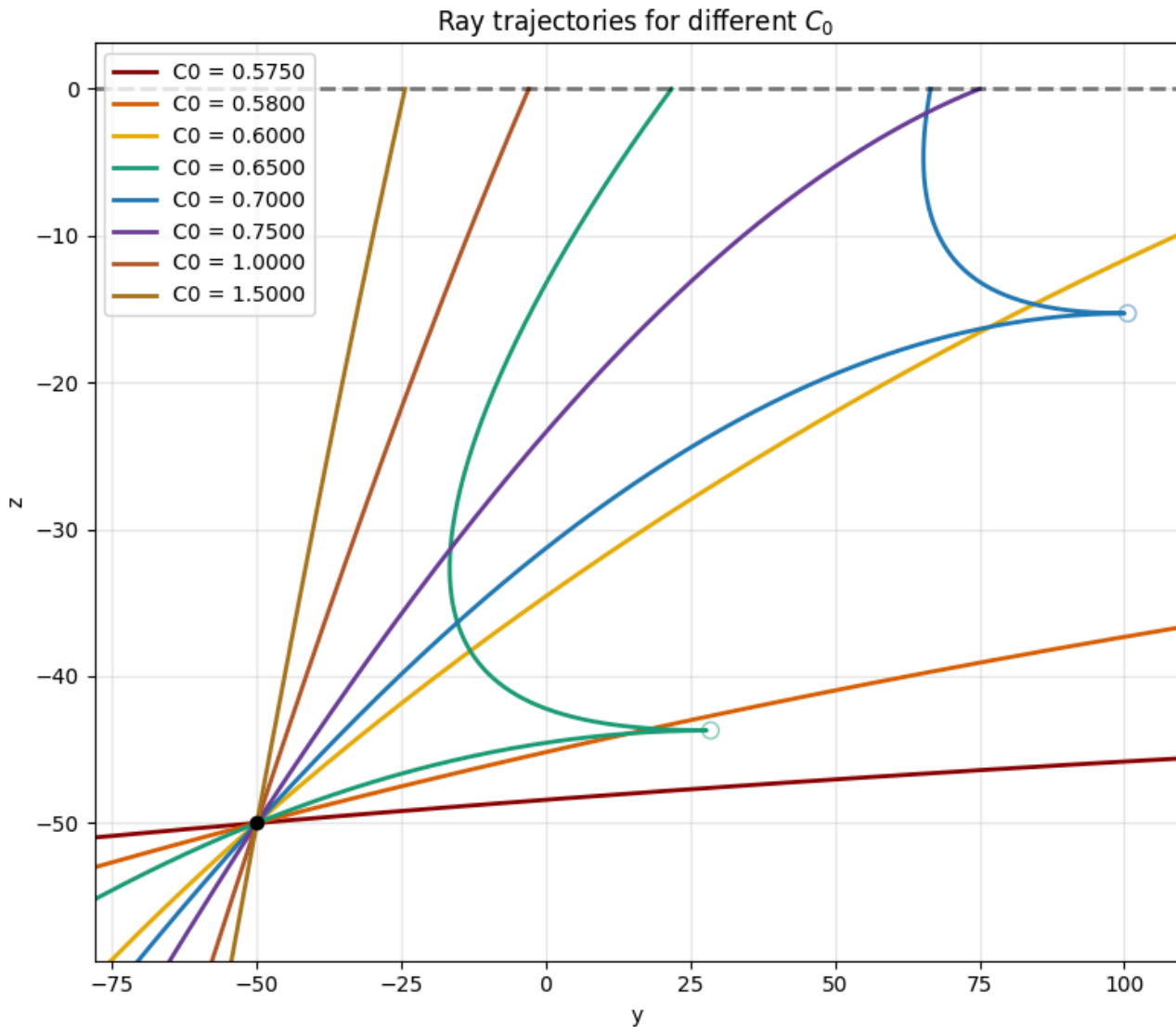
Choosing an initial point $\vec{x}_1 = (y_1, z_1)$ we just calculate the corresponding C_1 as follows:

$$C_1 = y_1 - Y(z_1, C_0)$$

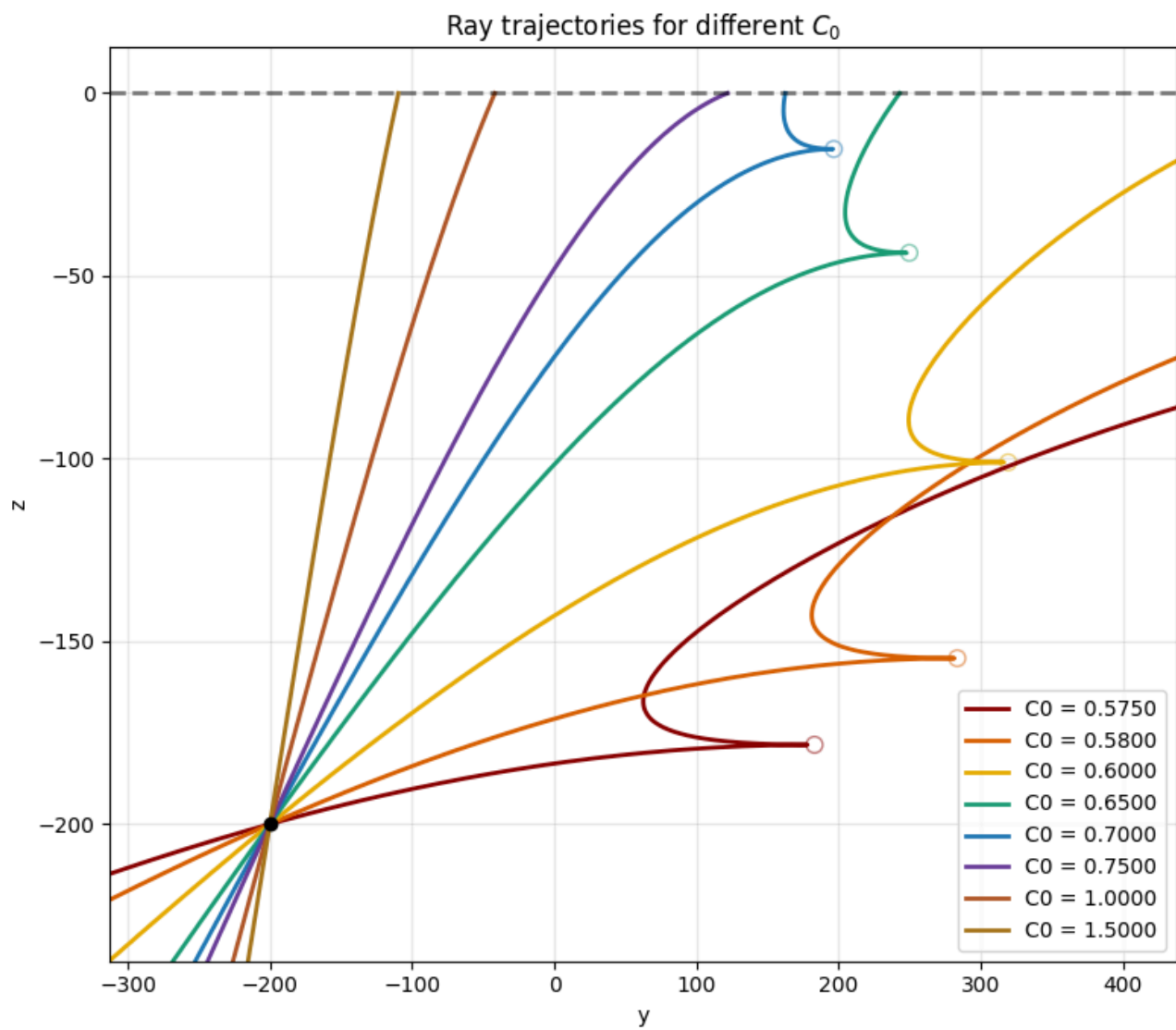
This allows us to find all the paths that a light ray which goes through a certain starting point $\vec{x}_1$ takes. Remember that this is only valid up to the point where the ray becomes horizontal and the solution has a singularity. The following plots show for different starting positions how the solutions look for different C_0 parameters. The turning points are highlighted with a small circle if visible.

First a shallow starting point at $\vec{x}_1 = (-50 \text{ m}, -50 \text{ m})$:

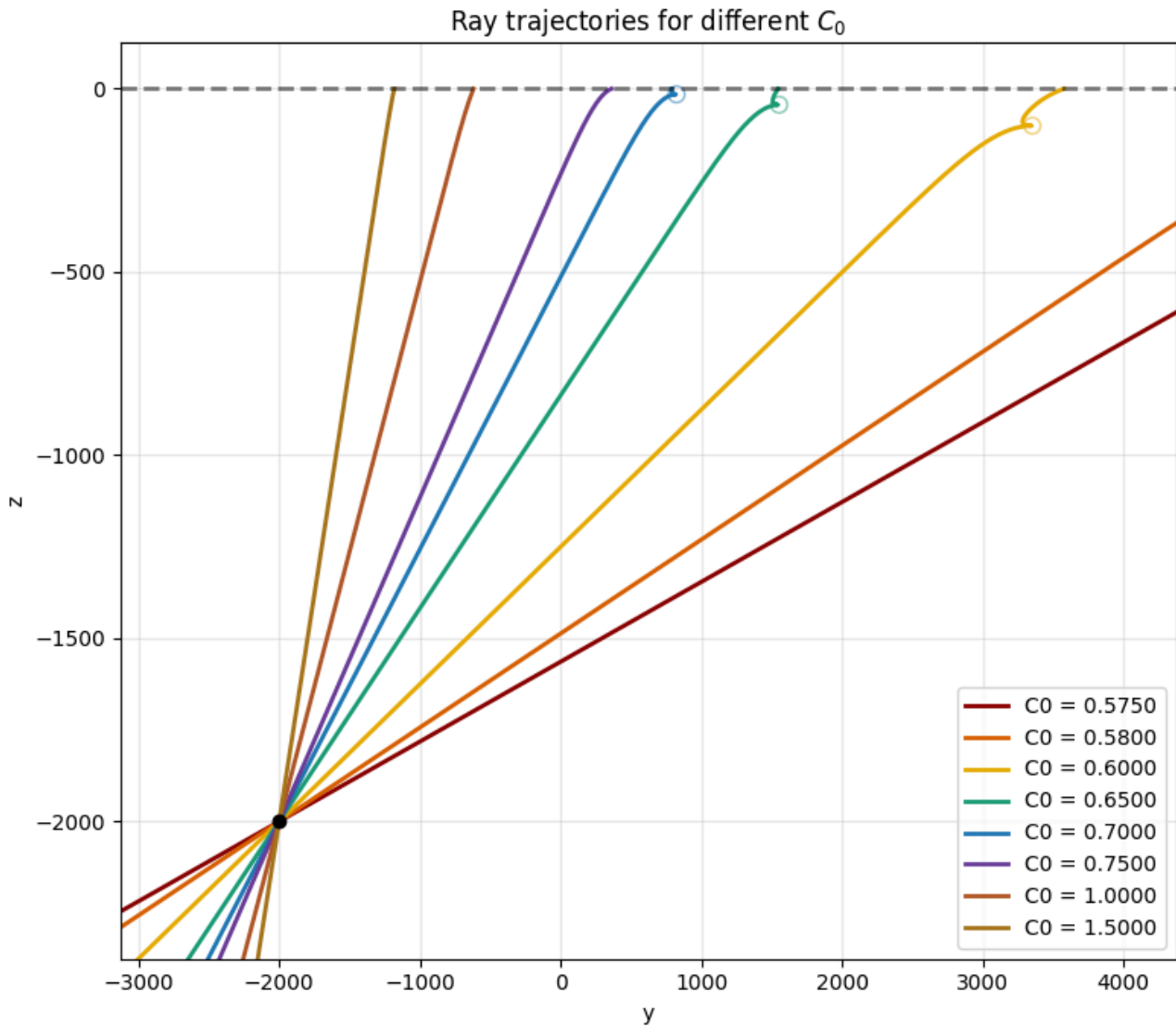
We have to be careful here: the lowest three solutions are not actually the physical paths but an upper branch solution of a too low C_0 . This behavior needs to be kept in mind when searching possible ray paths.



Then a slightly deeper $\vec{x}_1 = (-200 \text{ m}, -200 \text{ m})$:



And for a very deep starting point $\vec{x}_1 = (-2000 \text{ m}, -2000 \text{ m})$:

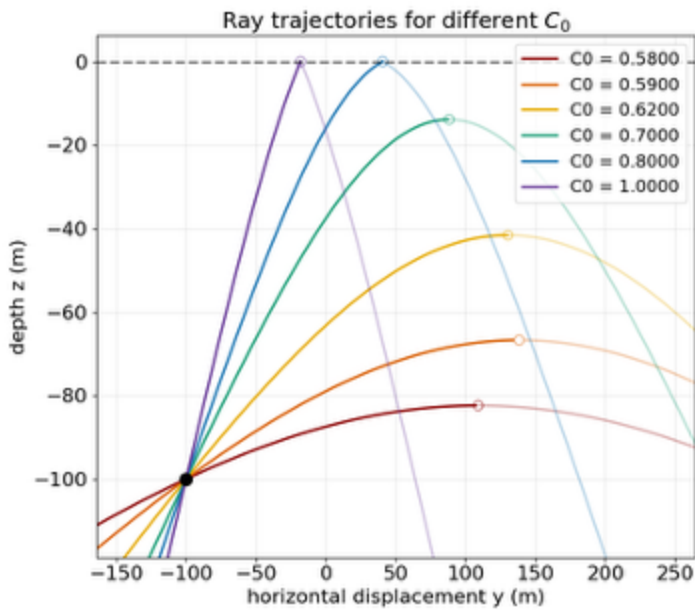
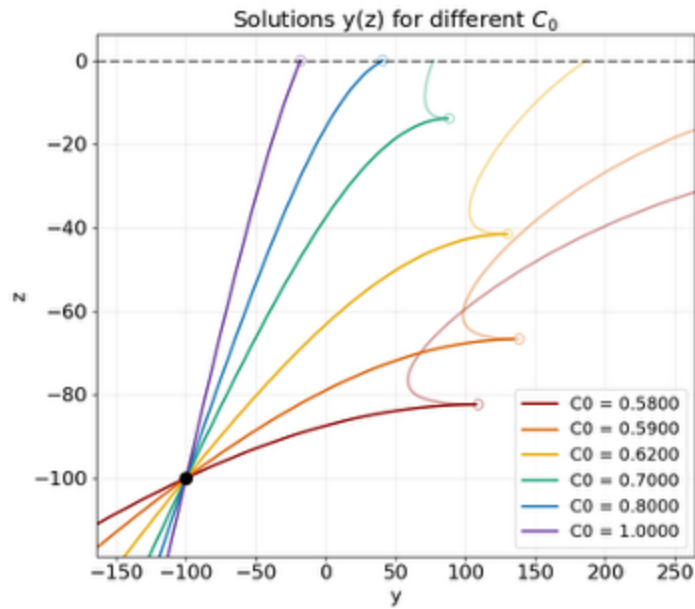


We can see that by sweeping through C_0 we can tune the ray path and look at different launch angles at the starting point.

Direct, Refracted and Reflected Rays

Light propagating upwards will bend towards the horizontal as described by the solutions shown above until the turning point. From there on it will bend further down again, symmetrically to the upwards propagation before the turning point. We can visualize this by mirroring the solution at the turning point. The following two images show first the path solutions and then the interpreted paths after being mirrored at the turning point or - if reached - the ice surface, corresponding to a reflection at the ice-air-boundary.

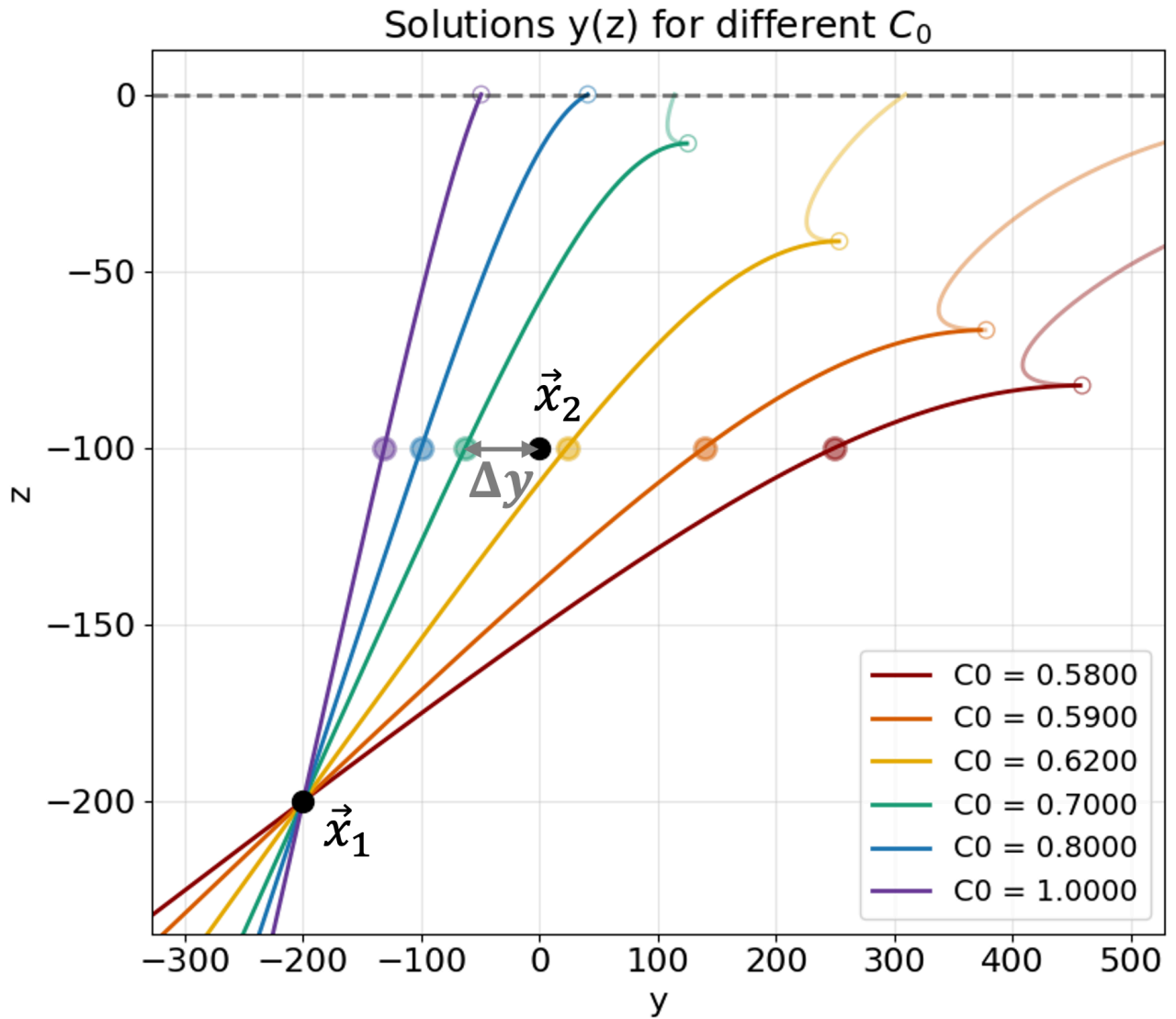
$\vec{x}_1 = (-100 \text{ m}, -100 \text{ m})$:



Numerical Solution Finding

Now that we are able to calculate the expected ray solution depending on the starting point $\vec{x}_1$, the ray parameter C_0 and the parameters of the exponential model, we can think about how to actually find the correct solution that connects two points (emitter: particle shower, receiver: antenna) of choice.

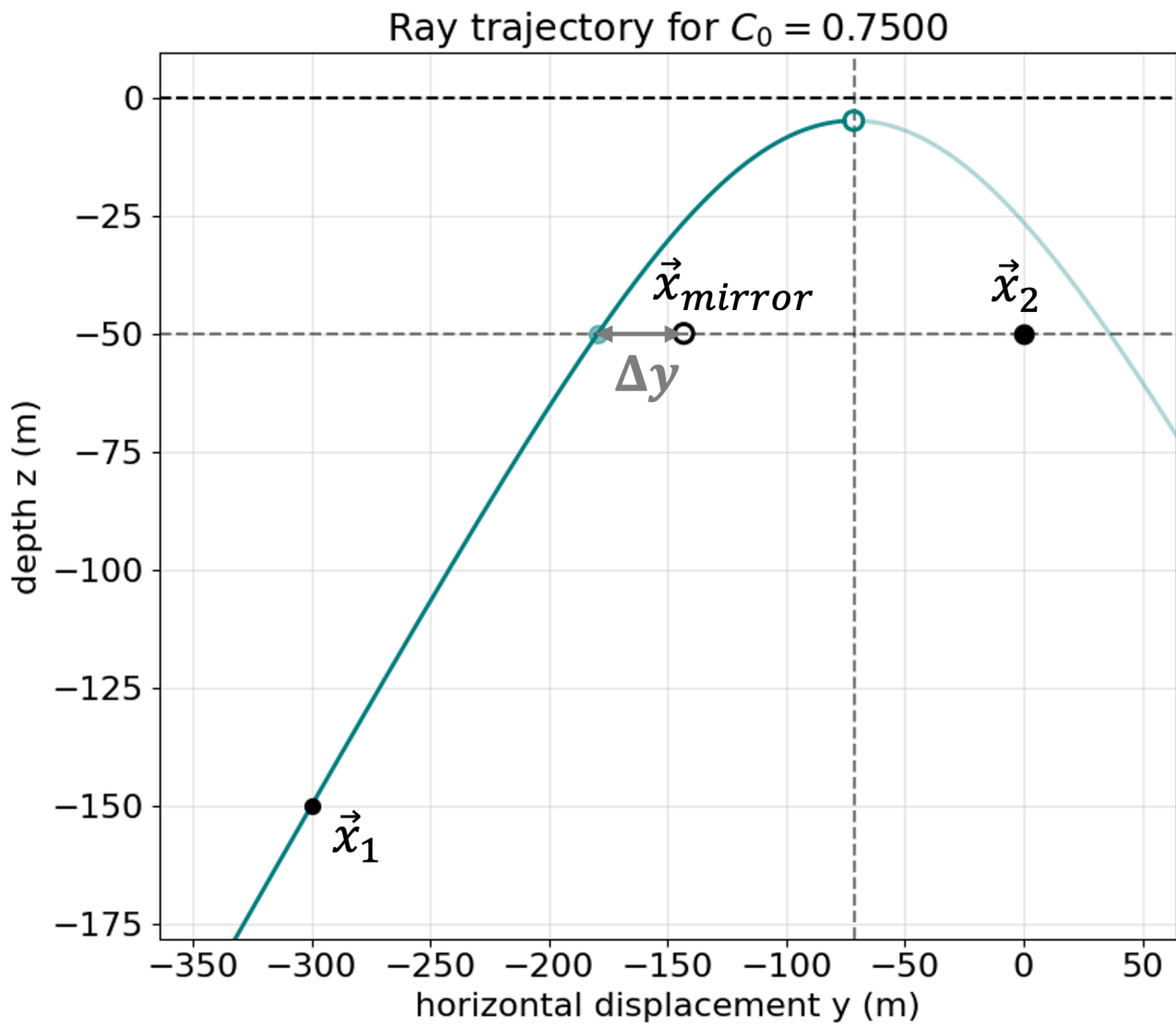
In the case of **Direct Rays** ($y_{turn} > y_2$) we evaluate the horizontal displacement $y(z_2)$ for a given C_0 at the depth of the receiver (z_2). Then we can calculate the horizontal difference to the antenna $\Delta y = y_2 - y(z_2)$.



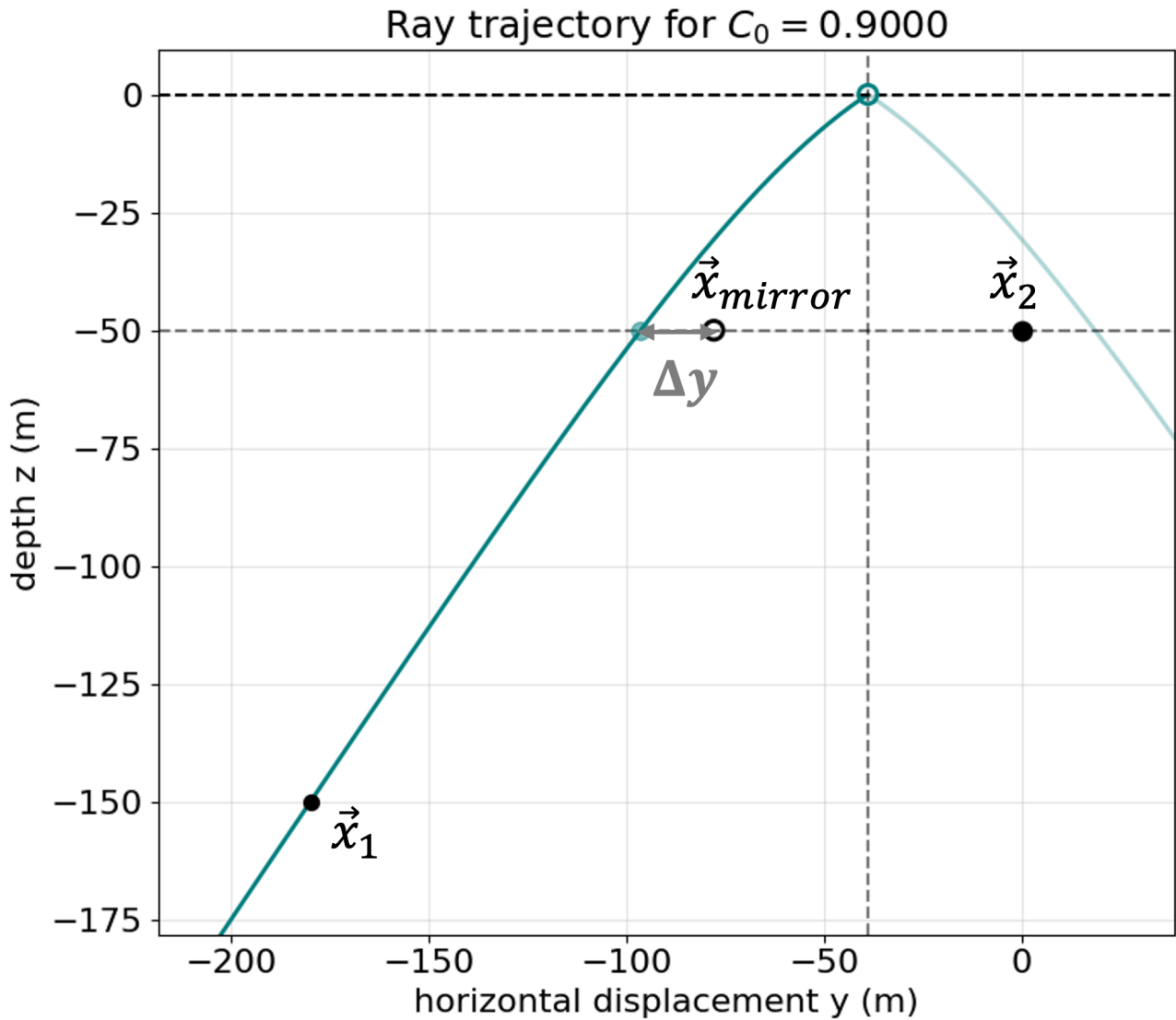
If we want to instead search the C_0 that creates a refracted ray which first reaches is turning point and then goes down again to our receiver, we can transform this to a direct ray problem again, by first calculating the turning point and then mirroring our target point at the y-value of the turning point:

$$y_{mirror} = y_2 - 2 \cdot (y_2 - y_{turn}) = 2 \cdot y_{turn} - y_2$$

This allows us to simply evaluate the rising ray path at the antenna depth z_2 and the value to minimize is instead the difference $\Delta y = y_{mirror} - y(z_2)$.



If we have a turning point above the ice surface, we just set the turning point to $y(z = 0.0)$ and proceed with the mirroring approach to get the rays that arrive at our receiver after a reflection at the surface, as shown in the next plot.



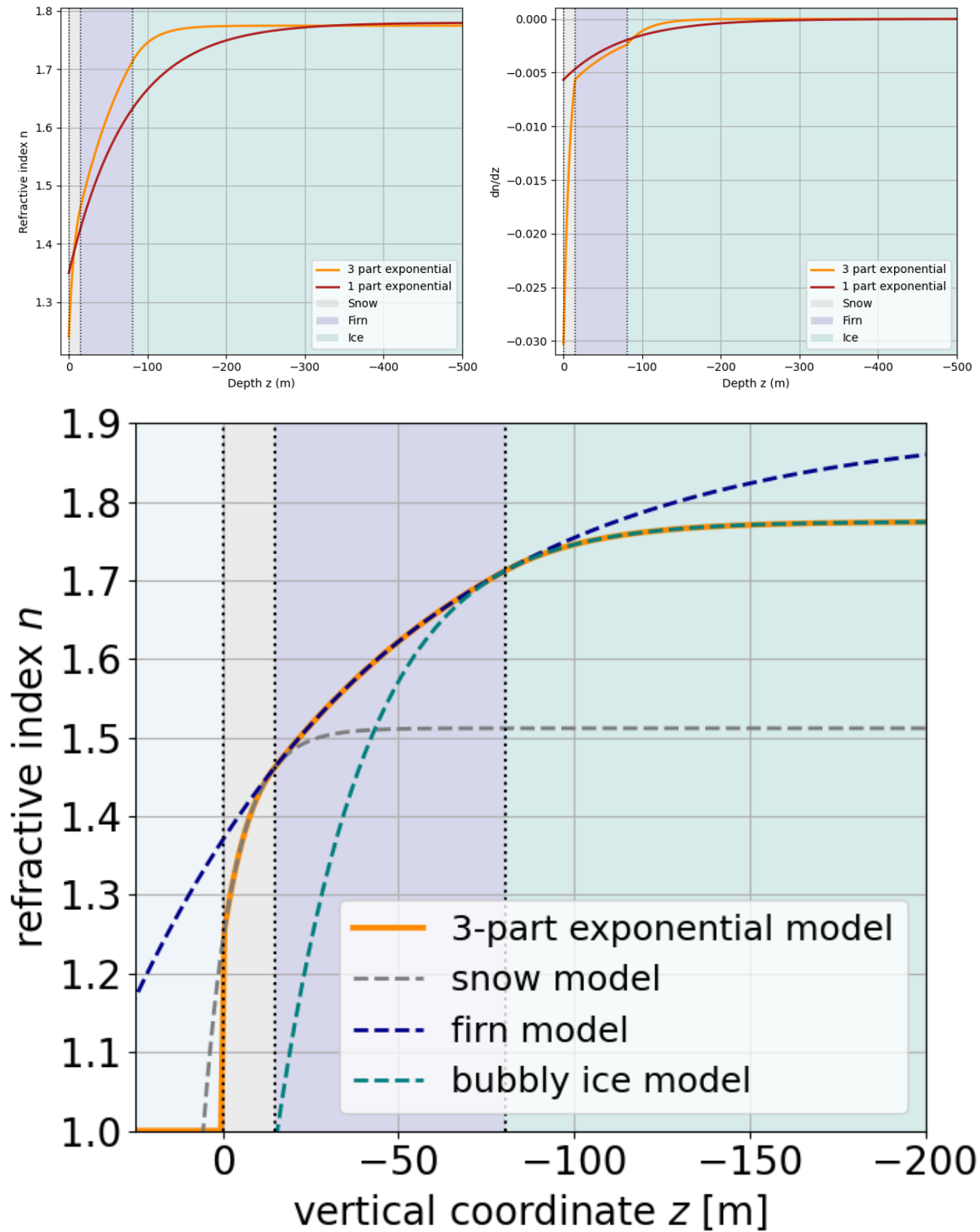
Unfortunately we can't directly calculate the ray parameter $C_0(x_1, x_2)$ that defines our ray tracing solution, so we instead have to minimize the differences Δy . The C_0 for which $\Delta y \approx 0$ characterizes a ray that goes to both of our points. For this we use different numerical minimization algorithms, that allow us to exploits how Δy varies with C_0 and we either find the root of $(\Delta y)^2$ with `scipy.optimize.root`, or find the zero point of Δy using `scipy.optimize.brentq`. While Brent's algorithm is generally faster, it is sensitive to the choice of the boundary values for C_0 , which is why we first use the stable root finder and then start again from the found solution to search for the second solution.

From a single exponential to a multi layer case

Measurements showed that the simple exponential model might not accurately enough represent the situation in real glacial ice. A better approximation could be found with a 3 layer exponential model as seen below. As for the single exponential model, the refractive index is defined in the form of

$$n(z) = n_{ice,i} - \Delta_{n,i} \exp(z/z_{0,i})$$

where different parameters are used for different regions of z . Those parameters are chosen to better depict how $n(z)$ changes in the shallow region.



Note that at layer boundaries between different exponential models the following holds:

$$n_1(z_b) = n_2(z_b)$$

$$\frac{dn_1}{dz}(z_b) = \frac{dn_2}{dz}(z_b)$$

Now if we go back again and have a look at the differential equation we found above:

$$\frac{\dot{y}}{\sqrt{\dot{y}^2 + 1}} n(z) = \frac{1}{C_0} = \text{constant}$$

This means that we still can characterize a ray path from the starting point $\vec{x}_1$ and C_0 . Now for each layer we need to find the correct offset $C_{1,i}$ so that we get the $y(z)$ with the physically correct ray path. The smooth transitions in the refractive index model ensures also a smooth ray path across the layer boundaries.

Let's set the starting point in an arbitrary layer i . Then we can calculate the $C_{1,i}$ for this layer by evaluating

$$C_{1,i} = y_1 - Y_i(z_1, C_0)$$

where Y_i denotes the function $Y(z, C_0)$ evaluated using the layer parameters corresponding to layer i . Now we can calculate ray position at boundary to the upper layer by evaluating $Y_i(z_b, C_0) + C_{1,i}$. We can then propagate this upwards layer to eventually find the correct C_1 for each of them. This is of course only possible for layers that are reached and the ray is not turning before.

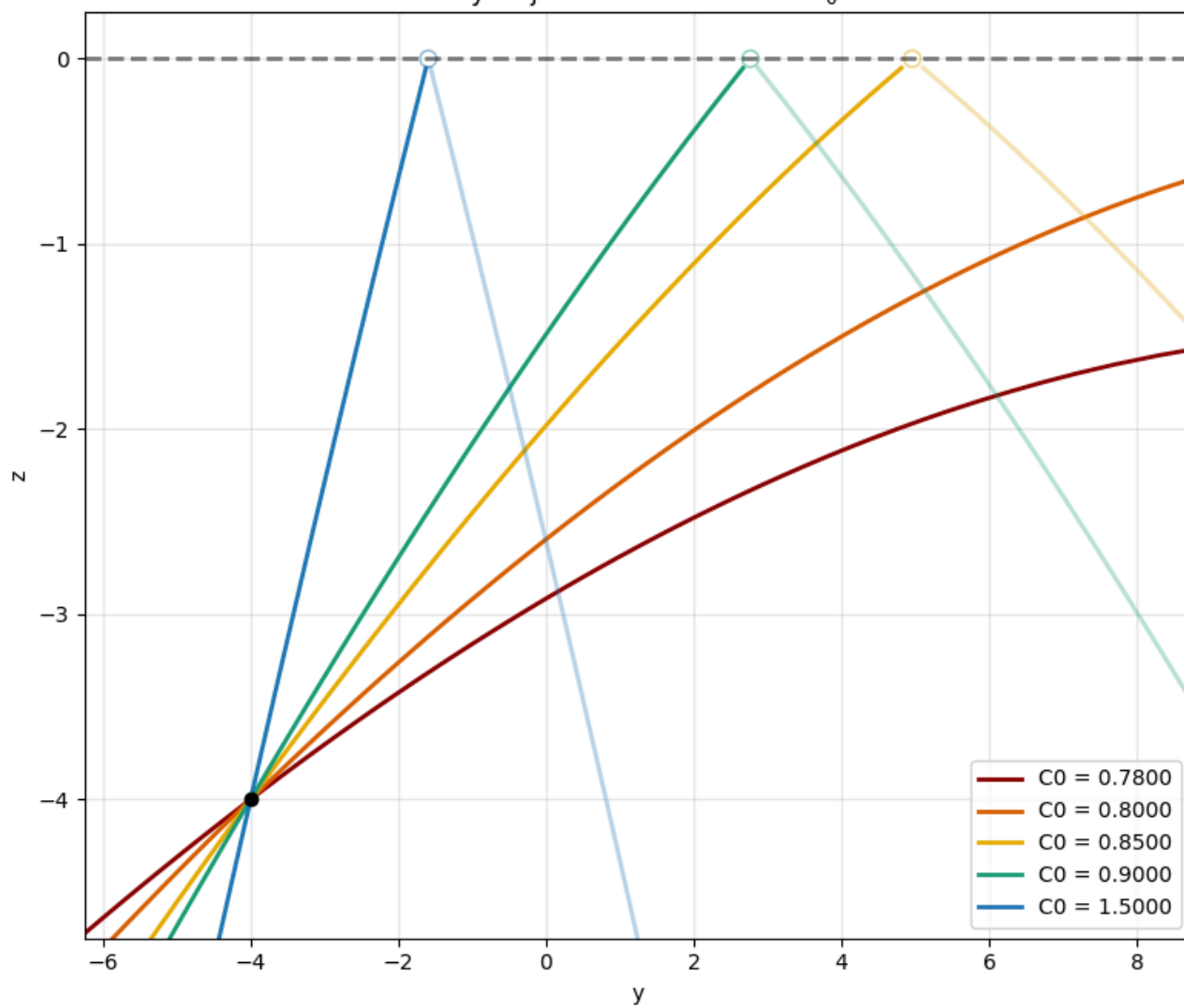
We end up with a vectorized description of $y(z)$ that allows us to calculate the solution to the whole "displacement-over-depth" problem in a way that is not that different than it is for the single layer exponential case:

$$y(z) = Y_i(z, C_0) + C_{1,i}$$

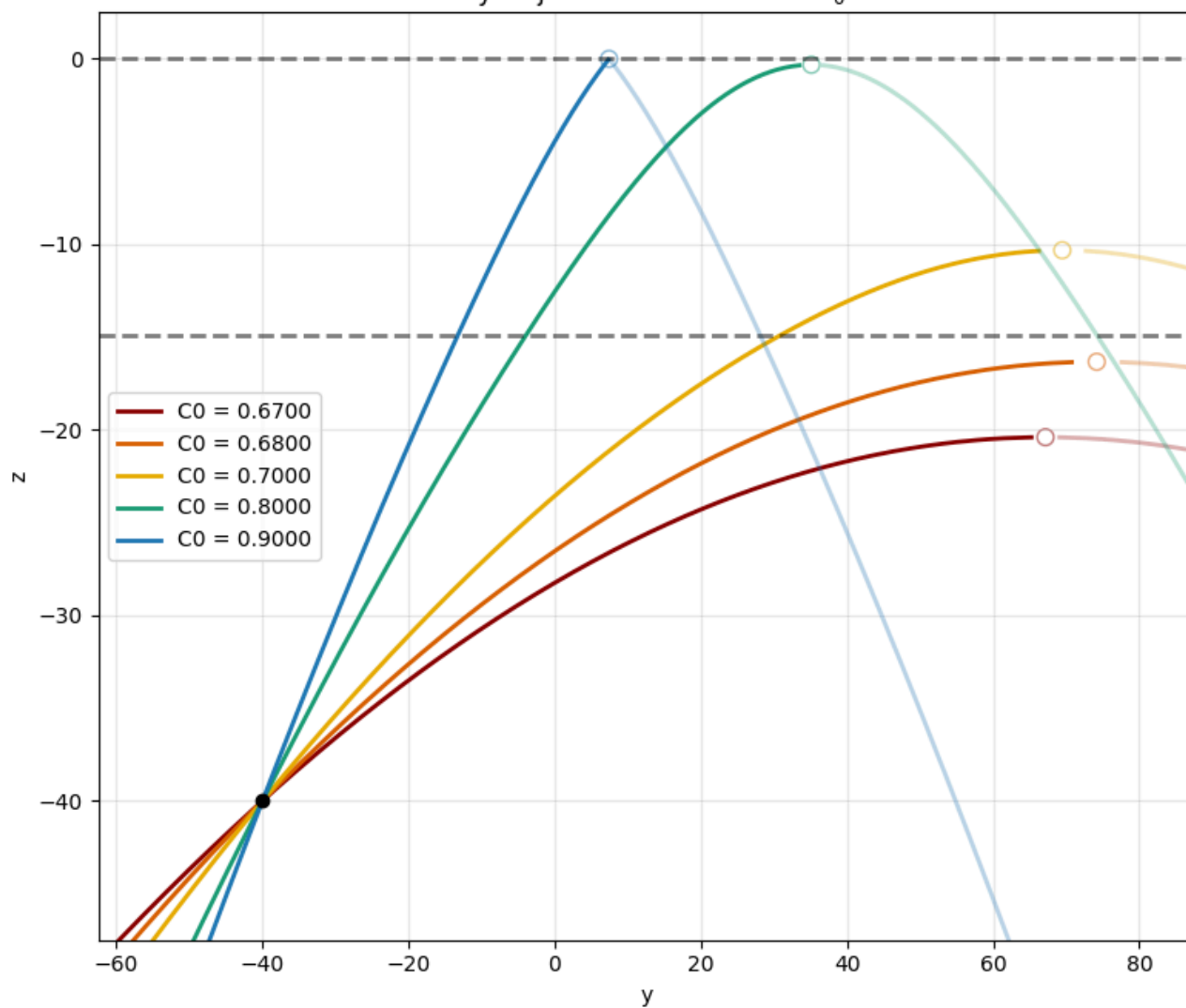
This reduces our problem again to the situation that we have to find the correct C_0 that connects two points of choice, which we now can tackle in the same way as it was done before by minimizing the difference between the evaluated $y(z_2)$ and y_2 at the target point $\vec{x}_2 = (y_2, z_2)$.

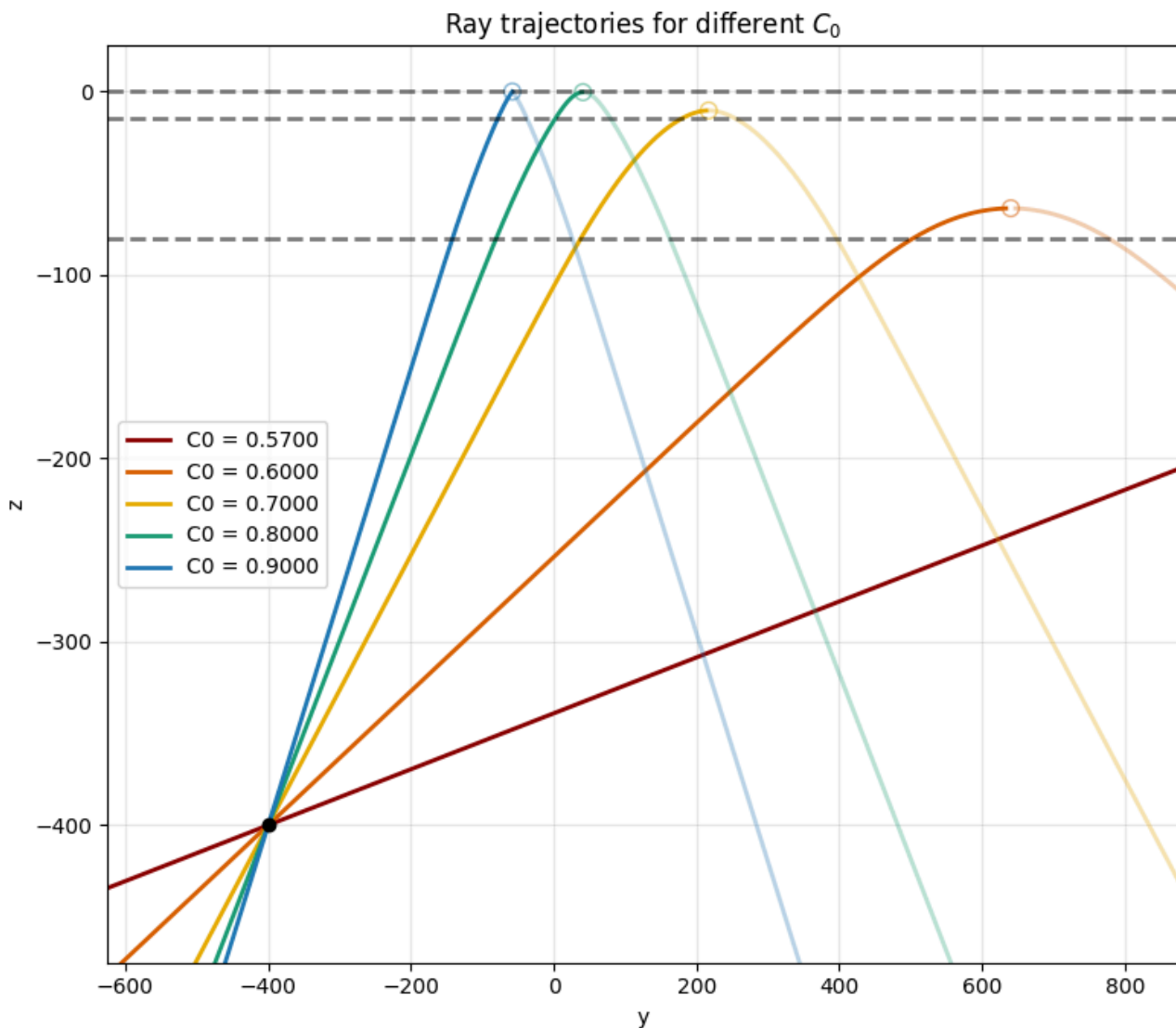
The following plots show the ray paths for different C_0 for the multilayer ice model `iop3`. The dashed lines show the layer boundaries. Notice that there is no jump in the refractive index there and no reflection is taking place (only on the ice surface at $z = 0$). These are just the regions where the corresponding exponential model is used.

Ray trajectories for different C_0



Ray trajectories for different C_0

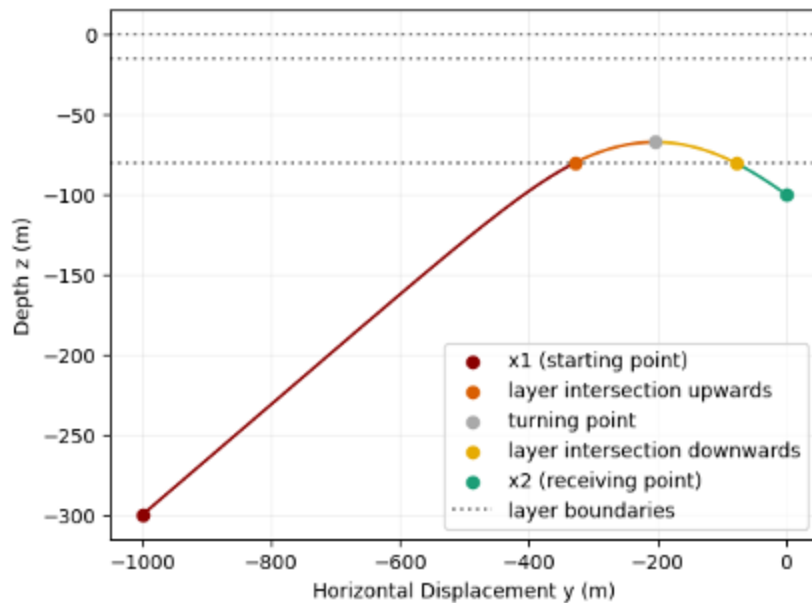




Calculating the path specific parameters (path length, travel time, focusing, ...)

Our goal will be to find and evaluate the signal paths for a given starting point (particle shower) to a receiver (antenna). We follow the same approach as in the single layer case to minimize Δy in an effective way using `scipy.root` and `scipy.brentq`. We can furthermore expand the analytical calculation of path length, travel time, focusing factor, launch and receiving angles as well as the numerical integration of the attenuation factor to our multi-layer case.

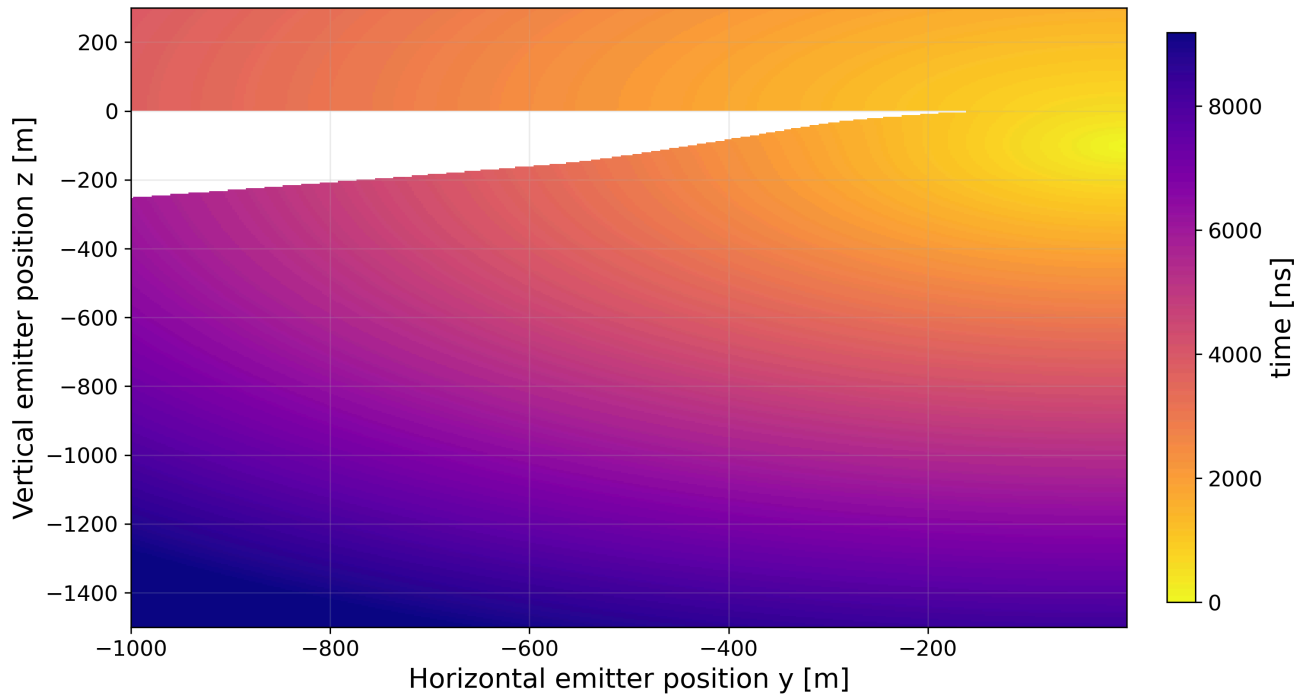
The first step is to split up the ray paths into monotonous segments that are contained within a single layer. Within each layer, the analytical solutions for path length, travel time and focusing factor are still valid and with the precomputed segments we can easily evaluate the integrals at the segment borders and then sum up (or multiply) each segments contribution.



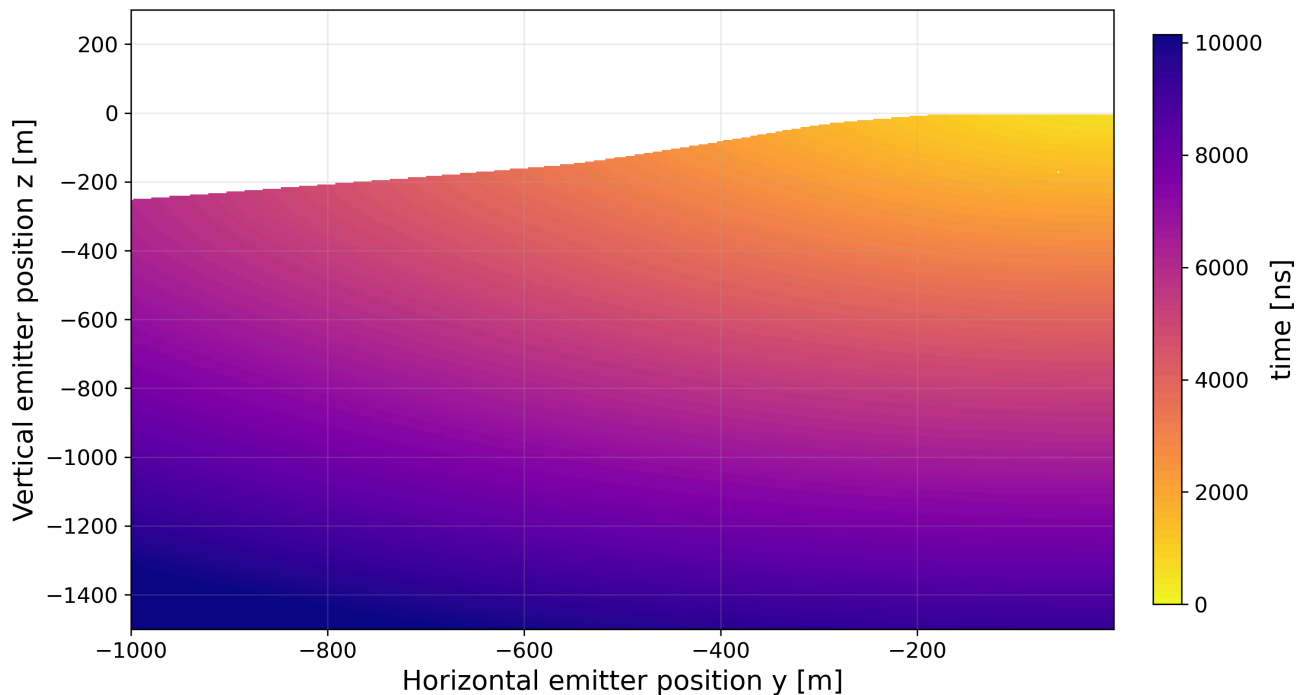
This allows us to efficiently calculate the path length, travel time and focusing factor analytically within each segment and then add the contribution from all segments. The analytical functions from the single layer analytic raytracer can be nicely reused. The equations for the path length and travel time are derived in Appendix C of <https://doi.org/10.1140/epjc/s10052-020-7612-8>. For deeper insights on the focusing factor see Appendix A of <https://open.fau.de/handle/openfau/37310>.

The following plot shows the travel time from each point to the receiver at $\vec{x}_2 = (0 \text{ m}, -100 \text{ m})$. First plot shows the lower C_0 solutions, so either the direct rays or - if we have two refracted solutions - the lower one of the two rays. The second plot shows the solutions with the higher C_0 so the steeper rays which get bent or reflected at the surface:

**Travel Time for Direct & Refracted Rays
multilayer - greenland_3exp_nils**



**Travel Time for Refracted & Reflected Rays
multilayer - greenland_3exp_nils**



Literature:

- „NuRadioMC: Simulating the Radio Emission of Neutrinos from Interaction to Detector“. *The European Physical Journal C* 80, Nr. 2 (2020): 77. <https://doi.org/10.1140/epjc/s10052-020-7612-8>.
- Bouma, Sjoerd. *Direction Reconstruction of Radio Signals in Neutrino Detectors in Ice*. 2025. <https://open.fau.de/handle/openfau/37310>

- Warnhofer et al. *What's new in NuRadioMC: Multilayer Analytic Raytracer*. Proceedings of Science, 11th International Workshop on Acoustic and Radio EeV Neutrino Detection Activities (ARENA2026). 2026.